ECE 310 Fall 2026
Lecture 7: z-transform Properties

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Learning Objectives

- Understand key properties of the z-transform and the region of convergence.
- Apply properties of the z-transform to derive the z-transform of discrete-time signals.



A Closer Look at the z-transform and ROC

We define the z-transform $X(z)$ for discrete-time signal $x[n]$ as

$$X(z) = \sum_{n-\infty}^{\infty} x[n] z^{-n}$$

where we must also denote all values of z where the above summation converges, known as the *region of convergence* (ROC).

$|z| > 2$

$z = a + jb = Re^{j\theta}$

$|z| = R > 2$



A Closer Look at the z-transform and ROC

It is important to note that $X(z)$ is not defined for values of z that lie outside the ROC even if the expression for $X(z)$ computes a finite value.

Consider

$$x[n] = \left(\frac{1}{2}\right)^n u[n] \overset{z}{\longleftrightarrow} X(z) = \frac{1}{1-\frac{1}{2}z^{-1}}, |z| > \frac{1}{2}$$

$z = \frac{1}{4}$

$$X(z)\Big|_{z=\frac{1}{4}} = \frac{1}{1-\frac{1}{2}\left(\frac{1}{4}\right)^{-1}} = \frac{1}{1-\frac{1}{2}(4)} = \frac{1}{-1} = -1 \; ?$$

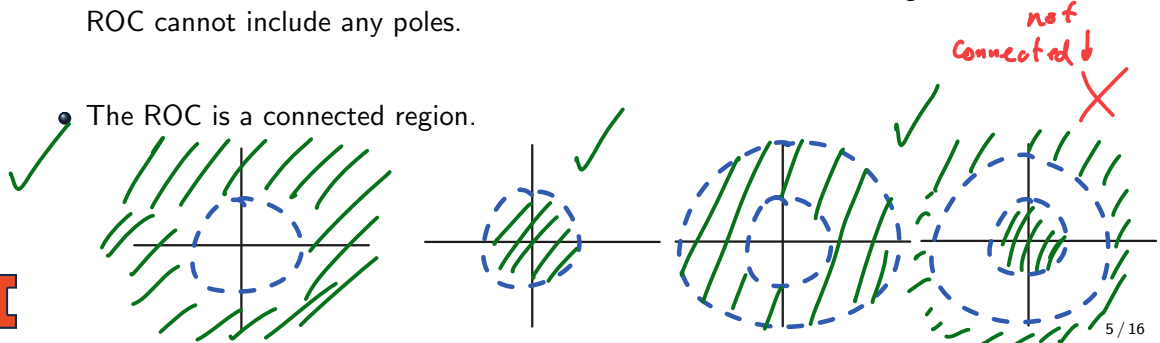
$$X(z)\Big|_{z=\frac{1}{4}} = \sum_{n=-\infty}^{\infty} x[n]\left(\frac{1}{4}\right)^{-n} = \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n u[n] \cdot \left(\frac{1}{4}\right)^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n (4)^n = \sum_{n=0}^{\infty} 2^n \to \infty !$$

A Closer Look at the z-transform and ROC

Let's summarize some important properties for the definition of the z-transform and region of convergence:

- Values of z where $X(z) = 0$ are referred to as *zeros* while values where $X(z) \to \infty$ are *poles*.
- The ROC defines values of z for which the z-transform sum converges. Thus, the ROC cannot include any poles.
- The ROC is a connected region.



A Closer Look at the z-transform and ROC

For finite-length signals, the ROC is the entire complex z-plane, except perhaps $z = 0$ for causal sequences or $z = \infty$ for non-causal sequences.

Ex: $x[n] = \delta[n] + \delta[n-2] \xleftrightarrow{\mathcal{Z}} 1 + z^{-2}$, ROC: all z except $z=0$

$= 1 + \frac{1}{z^2}$ → ROC: $z \neq 0$

Ex: $x[n] = \delta[n] + \delta[n+2] \xleftrightarrow{\mathcal{Z}} 1 + z^2$, ROC: all z except $z \to \infty$

ROC: $|z| \neq \infty$

$z \neq \pm\infty$

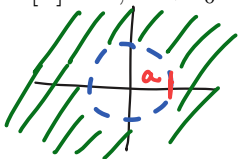
A Closer Look at the z-transform and ROC

For infinite-length signals, the ROC can have one of the following shapes: menti.com: 2105 8661

- Right-sided sequence, i.e. $x[n] = 0, n < n_0$: ROC $= |z| > a,\ a \geq 0.$

Ex: $X[n] = u[n-1]$

$\rightarrow n_0 = 1$



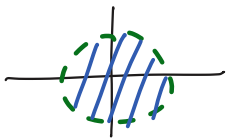
$a = |p_{max}|$

↖ pole with largest magnitude

- Left-sided sequence, i.e. $x[n] = 0, n > n_0$: ROC $= |z| < a,\ a \geq 0.$

Ex: $X[n] = u[-(n+3)]$

$n_0 = -3$

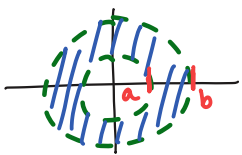


$a = |p_{min}|$

↖ smallest magnitude pole

- Two-sided sequence: ROC $= a < |z| < b,\ a \geq 0, b \geq 0, a < b.$

Ex: $X[n] = \left(\frac{1}{2}\right)^n u[n] + (3)^n u[-n-1]$



Example 1: Computing ROC

Determine the ROC of the below $X(z)$ assuming the corresponding $x[n]$ is right-sided. What is the ROC if $x[n]$ is left-sided?

$$X(z) = \frac{1 + 3z^{-1}}{1 + \frac{7}{4}z^{-1} - \frac{1}{2}z^{-2}}$$

* Note: there are poles and zeros @ $z = 0$ that cancel, thus, they have no effect on the ROC.

To find poles of $X(z)$, find roots of denominator

$1 + \frac{7}{4}z^{-1} - \frac{1}{2}z^{-2}$

a b c

$$\text{Roots} = \text{Poles} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-\frac{7}{4} \pm \sqrt{\frac{49}{16} + \frac{32}{16}}}{2}$$

$$= -\frac{7}{8} \pm \frac{\sqrt{\frac{81}{16}}}{2} = -\frac{7}{8} \pm \frac{9}{8} \rightarrow \text{poles @ } z = \frac{1}{4} \text{ and } z = -2$$

Right-sided (RS)

$|z| > |p_{max}| = 2$

$|z| > 2$

Left-sided (LS)

$|z| < |p_{min}| = \frac{1}{4}$

$|z| < \frac{1}{4}$

z-transform Properties: Time-shifting

Let $x[n]$ be a discrete-time signal with z-transform $X(z)$ and ROC R_x.

$$x[n-k] \overset{\mathcal{Z}}{\longleftrightarrow} z^{-k}X(z),\ k \in \mathbb{Z},\ \text{ROC} = R_x \text{ except possibly } z = 0 \text{ or } z = \infty$$

Ex: Let $x[n] = u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1-z^{-1}}$, $|z| > 1$

① $x[n-3] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{z^{-3}}{1-z^{-1}}$, $|z| > 1$ (pole @ $z=0$ is already outside ROC)

② $x[n+2] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{z^{2}}{1-z^{-1}}$, $1 < |z| < \infty$ (must exclude pole @ $z=\infty$)

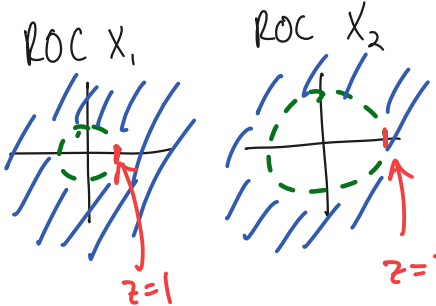
z-transform Properties: Convolution

Let $x_1[n]$ and $x_2[n]$ be discrete-time signals with z-transforms $X_1(z)$ and $X_2(z)$ and ROCs R_{x_1} and R_{x_2}, respectively.

$$x_1[n] * x_2[n] \overset{\mathcal{Z}}{\longleftrightarrow} X_1(z)X_2(z), \ \text{ROC} \ = \ \text{at least} \ R_{x_1} \cap R_{x_2}$$

intersection

there may be pole-zero cancellations



$$R_{x_1} \cap R_{x_2} = |z| > 3$$

z-transform Properties: Linearity

Let $x_1[n]$ and $x_2[n]$ be discrete-time signals with z-transforms $X_1(z)$ and $X_2(z)$ and ROCs R_{x_1} and R_{x_2}, respectively.

$$ax_1[n] + bx_2[n] \overset{\mathcal{Z}}{\longleftrightarrow} aX_1(z) + bX_2(z), \text{ ROC } = \text{ at least } R_{x_1} \cap R_{x_2}$$



z-transform Properties: Scaling

Let $x[n]$ be a discrete-time signal with z-transform $X(z)$.

$$a^n x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X\left(\frac{z}{a}\right), \; a \in \mathbb{C}, \; \text{ROC} \; = |a| R_x$$



Example 2: Applying z-transform Properties

Let $x_1[n] = \left(\frac{1}{2}\right)^n u[n] - c\left(\frac{1}{2}\right)^{n-1} u[n-1]$, $x_2[n] = (-1)^n u[n]$ for $c \in \mathbb{R}$, and $y[n] = x_1[n] * x_2[n]$. Determine $Y(z)$ and its ROC when (a) $c = 1$, (b) $c = -1$.

$a^n u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1-az^{-1}}, \quad |z| > |a|$

$$Y(z) = \frac{1-cz^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)\left(1+z^{-1}\right)}$$

$$Y(z) = X_1(z)X_2(z)$$

$$X_1(z) = \frac{1}{1-\frac{1}{2}z^{-1}} - \frac{cz^{-1}}{1-\frac{1}{2}z^{-1}} = \frac{1-cz^{-1}}{1-\frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}$$

$$X_2(z) = \frac{1}{1+z^{-1}}, \quad |z| > 1$$

(a) $c = 1$

$$Y(z) = \frac{1-z^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)\left(1+z^{-1}\right)}, \quad |z| > 1$$

(b) $c = -1$

$$Y(z) = \frac{1+z^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)\left(1+z^{-1}\right)} = \frac{1}{1-\frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}$$

Note how the pole-zero cancellation in part (b) expanded the ROC!

z-transform Properties: Differentiation

Let $x[n]$ be a discrete-time signal with z-transform $X(z)$.

$$nx[n] \overset{\mathcal{Z}}{\longleftrightarrow} -z\frac{dX(z)}{dz}, \ \text{ROC} \ = R_x$$



Example 3: Applying z-transform Properties

Let $x[n] = \left(\frac{1}{3}\right)^n u[n]$. Determine the z-transform $Y(z)$ and ROC of $y[n]$ below.

$$y[n] = 2^n n x[n]$$

$$X(z) = \frac{1}{1-\frac{1}{3}z^{-1}}, \quad |z| > \frac{1}{3}$$

Option 1:

- $y[n] = 2^n n x[n] = 2^n v[n]$
- $v[n] = n x[n]$
- $V(z) = -z \frac{dX(z)}{dz}$

$$= -z \frac{d}{dz}\left(\frac{1}{1-\frac{1}{3}z^{-1}}\right)$$

$$= -z\left((-1)(1-\tfrac{1}{3}z^{-1})^{-2}\left(\tfrac{1}{3}z^{-2}\right)\right)$$

$$= \frac{\frac{1}{3}z^{-1}}{(1-\frac{1}{3}z^{-1})^2}$$

- $Y(z) = V\left(\frac{z}{2}\right)$

$$= \frac{\frac{1}{3}\left(\frac{z}{2}\right)^{-1}}{\left(1-\frac{1}{3}\left(\frac{z}{2}\right)^{-1}\right)^2}$$

$$= \frac{\frac{2}{3}z^{-1}}{\left(1-\frac{2}{3}z^{-1}\right)^2}, \quad |z| > \frac{2}{3}$$

Example 3: Applying z-transform Properties

Let $x[n] = \left(\frac{1}{3}\right)^n u[n]$. Determine the z-transform $Y(z)$ and ROC of $y[n]$ below.

$$y[n] = 2^n n x[n]$$

$$X(z) = \frac{1}{1-\frac{1}{3}z^{-1}}, \quad |z| > \frac{1}{3}$$

Option 2:

- $y[n] = n 2^n x[n] = n g[n]$
- $g[n] = 2^n x[n]$
- $G(z) = \frac{1}{1-\frac{1}{3}\left(\frac{z}{2}\right)^{-1}}$

$$= \frac{1}{1-\frac{2}{3}z^{-1}}$$

- $Y(z) = -z\frac{dG(z)}{dz}$

$$= -z\frac{d}{dz}\left(\frac{1}{1-\frac{2}{3}z^{-1}}\right)$$

$$= -z\left((-1)\left(1-\frac{2}{3}z^{-1}\right)^{-2}\left(\frac{2}{3}z^{-2}\right)\right)$$

$$= \frac{\frac{2}{3}z^{-1}}{\left(1-\frac{2}{3}z^{-1}\right)^2}, \quad |z| > \frac{2}{3}$$

Summary of Key z-transform Properties

Property	Signal	z-transform	ROC
Time-shifting	$x[n-k]$	$z^{-k}X(z)$	R_x except $z=0$ or $z=\infty$
Linearity	$ax_1[n]+bx_2[n]$	$aX_1(z)+bX_2(z)$	At least $R_{x_1}\cap R_{x_2}$
Convolution	$x_1[n]*x_2[n]$	$X_1(z)X_2(z)$	At least $R_{x_1}\cap R_{x_2}$
Differentiation	$nx[n]$	$-z\frac{dX(z)}{dz}$	R_x
Conjugation	$x^*[n]$	$X^*(z^*)$	R_x
Time reversal	$x[-n]$	$X\left(z^{-1}\right)$	$1/R_x$
Scaling	$a^n x[n]$	$X\left(z/a\right)$	$\lvert a\rvert R_x$
Real-part	$\mathrm{Re}\{x[n]\}$	$(1/2)[X(z)+X^*(z^*)]$	At least R_x
Imaginary-part	$\mathrm{Im}\{x[n]\}$	$(1/2)[X(z)-X^*(z^*)]$	At least R_x

